

Decontamination is a difficult problem. We're working to improve all of the above procedures to ensure that benchmark data does not appear in the training data.

2.4 SWE-bench (Verified, Pro, and Multilingual)

SWE-bench (Software Engineering Bench) tests AI models on real-world software engineering tasks.

For the [SWE-bench Verified](#) variant, developed by OpenAI, models are shown 500 problems that have been verified by human engineers to be solvable. We also assessed the model on [SWE-bench Multilingual](#)⁸. Here, “multilingual” refers to different programming languages: this variant assesses models on their solutions to 300 problems in 9 different languages. We ran this evaluation with extended thinking turned off and a 200k context window. [SWE-bench Pro](#), developed by Scale AI, is a substantially more difficult set of 1,865 problems.

	SWE-bench Verified	SWE-bench Pro	SWE-bench Multilingual
Claude Opus 4.5 (64k thinking)	80.60%	51.60%	76.20%
Claude Opus 4.5 (no thinking)	80.90%	52.0%	76.20%

Table 2.4.A Results for the three variants of the SWE-bench evaluation. All scores are averaged over 5 trials.

2.5 Terminal-Bench

[Terminal-Bench](#), developed by researchers at Stanford University and the Laude Institute, tests AI models on real-world tasks within terminal or command-line environments.

We ran [Terminal-Bench 2.0](#) using the Terminus-2 harness, in the Harbor scaffold. Low resource constraints in Terminal-Bench tasks introduced flakiness of up to 13%, primarily due to containers OOM’ing. When encountering failures, before killing the pods, we increased resource limits by 2× for every model we benchmarked. This reduced infra-related errors to <1%. The reported score for GPT-5.1-Codex-Max in Table 2.3.1 above uses a different harness (Codex CLI) and hosting environment and was not reproducible by us due to the model not being publicly available.

With a 128k thinking budget, Claude Opus 4.5 achieved a score of 59.27%±1.34% with 1,335 trials. With a 64k thinking budget, it achieved a score of 57.76%±1.05% with 2,225 trials.

⁸ Yang, J., et al. (2025). SWE-smith: Scaling Data for Software Engineering Agents. arXiv:2504.21798. <https://arxiv.org/abs/2504.21798>